Titan. When we looked at it through the telescope, we weren't seeing its surface. We knew nothing about what the surface was like. We had no idea how far below the clouds the surface was.

So, by the early 1970s, as a kind of legacy from Huygens and his line of intellectual descent, we knew at least that Titan has a dense methane-rich atmosphere, and that it's probably enveloped by a reddish cloud veil or aerosol haze. But what kind of cloud is red? By the early 1970s my colleague Bishun Khare and I had been doing experiments at Cornell in which we irradiated various methane-rich atmospheres with ultraviolet light or electrons and were generating reddish or brownish solids; the stuff would coat the interiors of our reaction vessels. It seemed to me that, if methane-rich Titan had red-brown clouds, those clouds might very well be similar to what we were making in the laboratory. We called this material tholin, after a Greek word for "muddy." At the beginning we had very little idea what it was made of. It was some organic stew made by breaking apart our starting molecules, and allowing the atoms—carbon, hydrogen, nitrogen—and molecular fragments to recombine.

The word "organic" carries no imputation of biological origin; following long-standing chemical usage dating back more than a century, it merely describes molecules built out of carbon atoms (excluding a few very simple ones such as carbon monoxide, CO, and carbon dioxide, CO_2). Since life on Earth is based on organic molecules, and since there was a time before there *was* life on Earth, some process must have made organic molecules on our planet before the time of the first organism. Something similar, I proposed, might be happening on Titan today.

The epochal event in our understanding of Titan was the arrival in 1980 and 1981 of the *Voyager 1* and *2* spacecraft in the Saturn system. The ultraviolet, infrared, and radio instruments revealed the pressure and temperature through the atmosphere—from the hidden surface to the edge of space. We learned how high the cloud tops are. We found that the air on Titan is composed mainly of nitrogen, N_2, as on the Earth today. The other principal constituent is, as